Exercise 1 (SQL Fundametals)

1. SELECT *
FROM employees;

2. SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name,
salary
FROM employees
ORDER BY salary DESC;

first_name	last_name	~~department~~ salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53 000
David	White	52 000
Jessica	More	51 000
Laura	Hall	50 000
Jane	Smith	48 000

4. SELECT first_name,
last_name,
salary AS Highest_Paid
FROM employees
ORDER BY salary DESC
LIMIT 5;

first_name	last_name	Highest_Paid
Emily	Davis	62 000
Mike	Johnson	60000
Robert	Wilson	59 000
John	Doe	55000
Sarah	Brown	53 000
Daniel	Clark	53000

5. SELECT *
FROM employees
WHERE department = 'IT';

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

6. SELECT *
FROM employees
WHERE department = 'Finance' AND salary >58 000;

id	first_name	last_name	department	salary	hire_date	city
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston

7. SELECT first_name,
last_name,
department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	More	HR
~~Laura~~ Daniel	Clark	Marketing

8. SELECT first_name,
last_name,
department
FROM employees
WHERE NOT department = 'Finance';

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Sarah	Brown	IT
David	White	Marketing
Emily	Davis	IT
Jessica	Moore	HR
Daniel	Klark	Marketing
Laura	Hall	IT

9. SELECT first_name,
last_name,
department
FROM employees
WHERE department IN ('HR', 'IT', 'Finance')

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Mike	Johnson	Finance
Laura	Hall	IT

10.
```
SELECT first_name,
       last_name,
       department,
       salary,
       location
FROM employees
WHERE department = 'IT' AND salary > 50000 AND city = 'New York';
```

first_name	last_name	department	salary	location
John	Doe	IT	55 000	New York
Sarah	Brown	IT	53 000	New York

11.
```
SELECT first_name,
       last_name,
       department, salary
FROM employees
WHERE department IN ('Finance', 'Marketing') AND
      salary > 52 000
ORDER BY salary DESC;
```

first_name	Last_name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

12. SELECT DISTINCT city,
FROM employees
WHERE department NOT IN ('IT', 'HR');

City
Los Angeles
San Francisco
Houston
Chicago

13. SELECT first_name,
last-name,
department,
salary ,
hire_date
FROM employees
WHERE department <> 'Finance' AND salary >50000
ORDER BY hire_date ASC;

first-name	last-name	department	salary	hire-date

14. SELECT first_name,
last_name,
city,
department
FROM employees
WHERE city IN ('Chicago', 'Los Angeles') AND department
IN ('IT', 'Marketing')
LIMIT 3;

first_name	last_name	city	department
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago